Is my baby too thin?
Early growth and nutrition

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Clinical Professor

& Consultant (Paed)

Learning Objectives

1. Key determinants of growth and growth standards
2. Identify the causes of failure to thrive
3. Understand early nutrition and gastrointestinal adaptation
4. Appreciate the importance of breast feeding

Is this baby too thin?

Is this baby too thin?

eed proper
bedside
xamination
Marked wastin
over buttock

How do you know this baby is thin?

Important growth parameters/dimensions:

- Body proportion (compare with self)
- Growth record (compare with past)
- Comparison with other babies

Early growth – key concept

- Relative to peers (age- and sex- appropriate)
- Relative to other growth parameters (weight/height; +/- head circumference)
- Relative to previous growth parameters

THE LANCET

Advancing Early Childhood Development: from Science to Scale 2

Nurturing care: promoting early childhood development

*Pia R Britto, Stephen J Lye, Kerrie Proulx, Aisha K Yousafzai, Stephen G Matthews, Tyler Vaivada, Rafael Perez-Escamilla, Nirmala Rao, Patrick Ip, Lia C H Fernald, Harriet MacMillan, Mark Hanson, Theodore D Wachs, Haogen Yao, Hirokazu Yoshikawa, Adrian Cerezo, James F Leckman, Zulfiqar A Bhutta, and the Early Childhood Development Interventions Review Group, for the Lancet Early Childhood Development Series Steering Committee**

The UN Sustainable Development Goals provide a historic opportunity to implement interventions, at scale, to promote early childhood development. Although the evidence base for the importance of early childhood development has grown, the research is distributed across sectors, populations, and settings, with diversity noted in both scope and focus. We provide a comprehensive updated analysis of early childhood development interventions across the five sectors of health, nutrition, education, child protection, and social protection. Our review concludes that to make interventions successful, smart, and sustainable, they need to be implemented as multi-sectoral intervention packages anchored in nurturing care. The recommendations emphasise that intervention packages should be applied at developmentally appropriate times during the life course, target multiple risks, and build on existing delivery platforms for feasibility of scale-up. While interventions will continue to improve with the growth of developmental science, the evidence now strongly suggests that parents, caregivers, and families need to be supported in providing nurturing care and protection in order for young children to achieve their developmental potential.

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Original Investigation | Global Health

Association of Early Nutritional Status With Child Development in the Asia Pacific Region

Frederick K. Ho, PhD; Nirmala Rao, PhD; Keith T. S. Tung, MPH; Rosa S. Wong, PhD; Wilfred H. S. Wong, PhD; Joanna Y. L. Tung, MPH; Gilbert T. Chua, MBBS; Winnie W. Y. Tso, MBBS; John Bacon-Shone, PhD; Ian C. K. Wong, PhD; Aisha Yousafzai, PhD; Charlotte Wright, PhD; Patrick Ip, MPH, MBBS

Key Points

Question Is early childhood development associated with nutritional and body composition indicators?

Findings In this cross-sectional study of 7108 children, better early child development was significantly and linearly associated with height-for-age and low body mass index-for-age. After adjustment for body mass index and height, better development was associated with low mid-upper arm fat proportion but not with fat area.

Meaning These findings suggest that future studies should consider using height-for-age, body mass index-for-age, and body fat proportion to estimate the prevalence of child underdevelopment, and nutritional trials should examine to what extent the associations are causal.

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JAMA Network | Open

Original Investigation | Global Health

Weight-for-Height, Body Fat, and Development in Children in the East Asia and Pacific Region

Fanny Petermann-Rocha, PhD; Nirmala Rao, PhD; Jill P. Pell, MD; Carlos Celis-Morales, PhD; Ian C. K. Wong, PhD; Frederick K. Ho, PhD; Patrick Ip, MPH, MBBS

Key Points

Question What is the combined association of weight-for-height *z* score and body fat with early development in the East Asia and Pacific region?

Findings This cross-sectional study of 6815 children found that compared with children with normal weight and normal body fat, children with low body fat, either with wasting or normal weight, were more likely to have poor total, cognitive, language, and socioemotional development independent of confounding factors.

Meaning These findings reinforce the potential to use body fat, in addition to weight-for-height *z* score, to identify children who are not on track developmentally.

Child Growth

- Foundation of health and development
- A dynamic and interactive process
- Relative to norms

Key determinant of growth

- Depends on stage of life and age of child
- Fetal / Infancy – *Nutrition
- Primary school - *Hormone(s)

Key milestones of child growth

- Birth weight - 3.2-3.4 kg

- Weight – double by 4 month,
  triple by 10 month

- Height – ½ adult height by 3 yr,
  ¾ adult height by 9 yr

- Head Circumference – 85% adult HC by 3 yr

Child Growth Indicators

- Underweight
- Stunting
- Wasting

- Failure to thrive

Definition of Z-Score

- Statistical measurement of a score's relationship to the mean in a group of scores
- Z-score of 0 means the score is same as mean
- A Z-score can be positive or negative, indicating whether it is above or below the mean and by how many standard deviations

WHO Global Database on Child Growth and Malnutrition

Underweight -

a Z-score cut-off point of <-2 SD to classify low weight-for-age

WHO Growth Standards (global data / growth charts)

- www.who.int/childgrowth/standards/en/

WHO Global Database on Child Growth and Malnutrition

Stunting -

a Z-score cut-off point of <-2 SD to classify low height-for-age

Stunting rate under 5-year-old is a key health indicator

WHO Global Database on Child Growth and Malnutrition

Wasting -

a Z-score cut-off point of <-2 SD to classify low weight-for-height

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EAST ASIA AND THE PACIFIC

Validation, Finalization and Adoption of the East Asia-Pacific Early Child Development Scales (EAP-ECDS)

Nirmala Rao, Jin Sun, Marie Ng, Yvonne Becher, Diana Lee, Patrick Ip and John Bacon-Shone

EARLY CHILDHOOD DEVELOPMENT IN EAST ASIA AND PACIFIC

Culturally sensitive testing

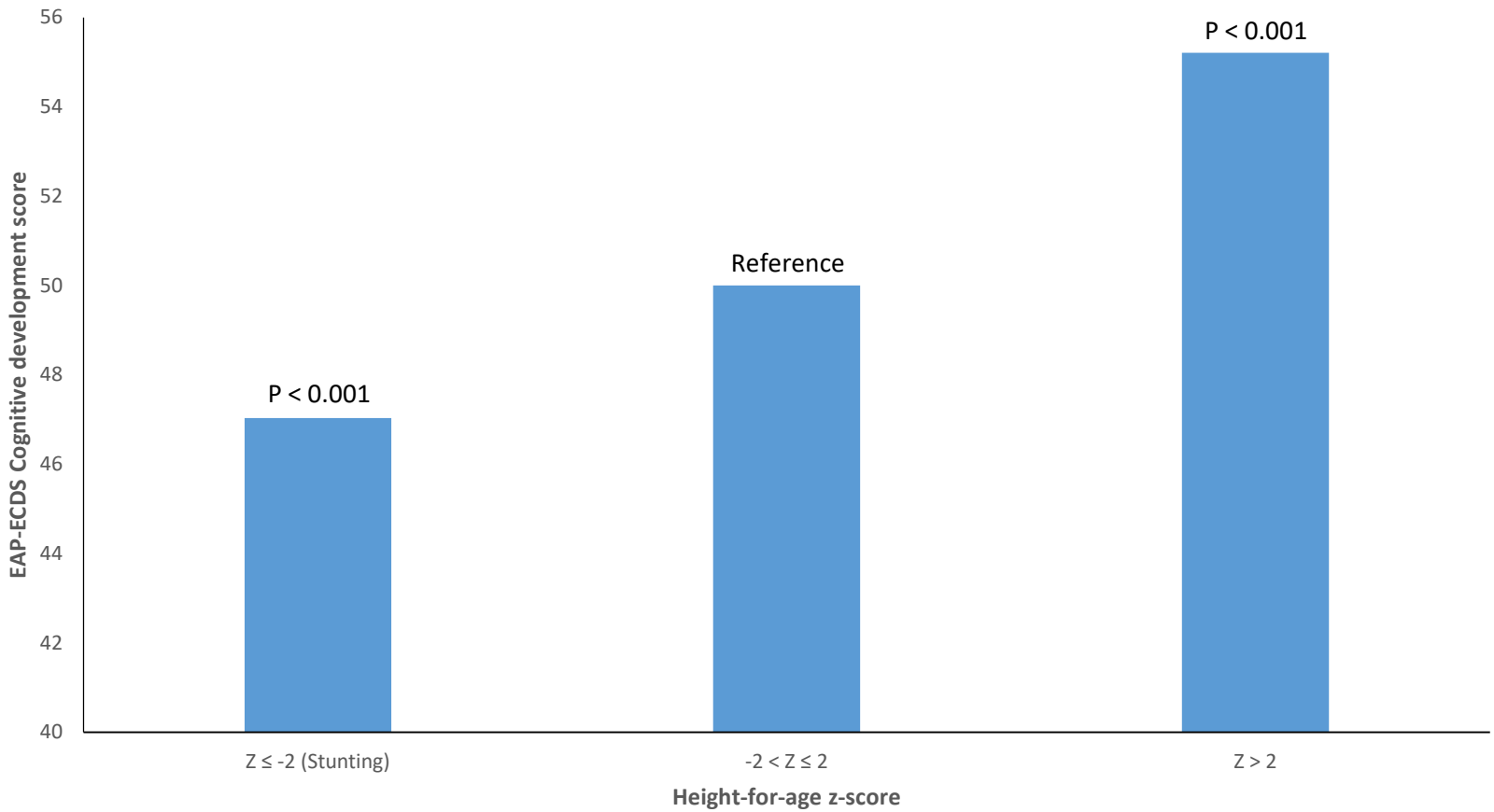
Monitoring early childhood development is clearly vital and yet there are at present no globally accepted tests that can be applied in all of the world's regions. In part this is due to concerns that assessment tools conceived in developed Western countries may not be appropriate in some cultural contexts. Yet the need for such testing is most acute in countries where a large proportion of children often suffer poor developmental outcomes and where government investment in early childhood development is limited.

To this end, the East Asia and Pacific Early Child Development Scales (EAP-ECDS) have been conceived and trialled over the period since 2010 – the first such assessment tool to have been originated, developed and validated specifically with children from the East Asia and the Pacific region in mind. Following an initial piloting in three countries, Cambodia, China, Mongolia, Papua New Guinea, Timor-Leste and Vanuatu participated in the

PANEL COMMENT

This large-scale research project aimed to provide Asian countries with a standard and reliable measurement tool to assess the holistic development of children from 3 to 5 years. This research is thoroughly conceptualized and directly engages with UNICEF's core work. There is important potential for policy impact and development of policy tools.

Stunting (height-for-age) and Cognitive development



Failure to Thrive - definition

- Failure of expected growth in children younger than 3 years

- Downward crossing of two percentile lines in weight over 6 months

Failure to Thrive - causes

Processing of nutrition:

- Inadequate intake
- Abnormal digestion/absorption
- Inability to utilize
- Excessive Loss

Increase in calorie requirement

Inadequate calorie intake (maternal factor)

- Failed breast feeding
- Wrong formula
- Poor preparation (misconception/tradition)
- Inappropriate feeding technique

Inadequate calorie intake (baby's factor)

- Congenital anomalies e.g. cleft palate
- CNS disorders – swallowing problem
- Distress – due to cardiopulmonary conditions
- GI – vomiting, GER

Malabsorption

- **Primary –**
  Uncommon (e.g. cystic fibrosis)

- **Secondary –**
  Post-gastroenteritis (common)
  Necrotising enterocolitis (NEC)
  Short gut syndrome
  Food allergy/intolerance

Defective use of calories

- Genetic diseases (syndromal)

- Metabolic disorders –
  inborn errors of CHO metabolsim,
  aminoacidopathies

Process involved in nutrition uptake

- Coordinated sucking and swallowing
- Gastric emptying
- Intestinal motility
- Secretions:
  salivary, gastric, pancreatic, hepatobiliary

Process involved in nutrition uptake

- Enterocyte function: enzyme synthesis, absorption, mucosal protection
- Metabolism of products of digestion & absorption
- Expulsion of undigested waste products

Process involved in nutrition uptake: in-utero

- Fetal GI tract is exposed to constant passage of fluid that contains a range of physiologically active factors:
  - growth factors
  - hormones
  - enzymes
  - Immunoglobulins

- These play a role in mucosal differentiation and GI development as well as development of swallowing and intestinal motility

Process involved in nutrition uptake: at birth

- Gut of the newborn baby: the task of passing, digesting, and absorbing large quantities of intermittent boluses of milk

Process involved in nutrition uptake: Gut Hormones

- Gastrointestinal peptides: found in venous cord blood at birth in levels ~ to those of fasting adults

- In fetal distress a number of gut peptides are elevated which might account for passage of meconium (meconium stained liquor; MSL)

- With enteral feeding levels of gut hormones (motilin, neurotensin, GIP (gastric inhibitory peptide), gastrin, enteroglucagon, pancreatic polypeptide, rise rapidly

Gut hormones - influenced by

- Choice of Breast or Formula feeds
- Enteric intake (induces epithelia hyperplasia & stimulates microvillous enzymes production)
- Early enteral feeding (enteral feeding is strongly encouraged to promote GI function & differentiation)

Possible Roles for Gut Hormones in early infancy

Motilin	Increased gut motility
Enteroglucagon	Tropic to gut mucosa
Enteroglucagon, gastrin, pancreatic polypeptides	Intestinal mucosal and pancreatic growth
Gastric Inhibitory polypeptide (GIP)	Stimulus to insulin release

Pancreas

- Pancreatic function: rel. deficient at birth, mature levels of pancreatic enzymes are not achieved until late infancy

- Pancreatic amylase activity increases after 4-6 months
- Lipase levels do not approach adult efficiency until about 6 months

Protein Digestion

Factor	In Early Infancy Compared to Adult levels
Gastric Acid	Lower production
Trypsin	Activity reduced
Chymotrypsin	Low levels
Pancreatic Proteases	Low levels
Intestinal Mucosal peptidases	Adequate

Carbohydrate Digestion

Factor	In Early Infancy Compared to Adult levels	Compensating Mechanisms
Salivary Amylase	Lower levels	Stays active in stomach
Pancreatic amylase	Very low levels	Breastmilk amylase
Disacharidases	Adequate levels	Fermentation and absorption in large intestine

Fat Digestion

Factor	In Early Infancy Compared to Adult levels	Compensating Mechanisms
Pancreatic Lipase	Very low levels	Lingual, gastric and breastmilk
Bile Acids	Low levels	Bile salt stimulated lipase

Motility - Upper GI

- Esophageal motility is decreased in the newborn
- Lower esophageal spincter is primarily above the diaphragm
- LES pressure is less for first months
- Gastric Emptying may be delayed

Motility - Intestinal

- Intestinal motility is more disorganized
- Prolonged transit time in upper intestines may improve absorption of nutrients
- Rapid emptying of ileum and colon may reduce time for water and electrolyte absorption and increase risk of dehydration

Maturation in First Year

- LES tone increases after 6 months and is associated with less reflux in most infants
- Gastric acid and pepsin activity don't reach adult levels until 2 years
- Pancreatic amylase increases by 6 months
- Retention of lactase activity is typical until 3 to 5 years
- Fat absorption does not approach adult efficiency until about 6 months
- Lipase reaches adult levels by 2 years

Kidney

- Limited ability to concentrate urine in first year due to immaturities of nephron and pituitary

- Potential renal solute load determined by nitrogenous end products of protein metabolism, sodium, potassium, phosphorus, and chloride

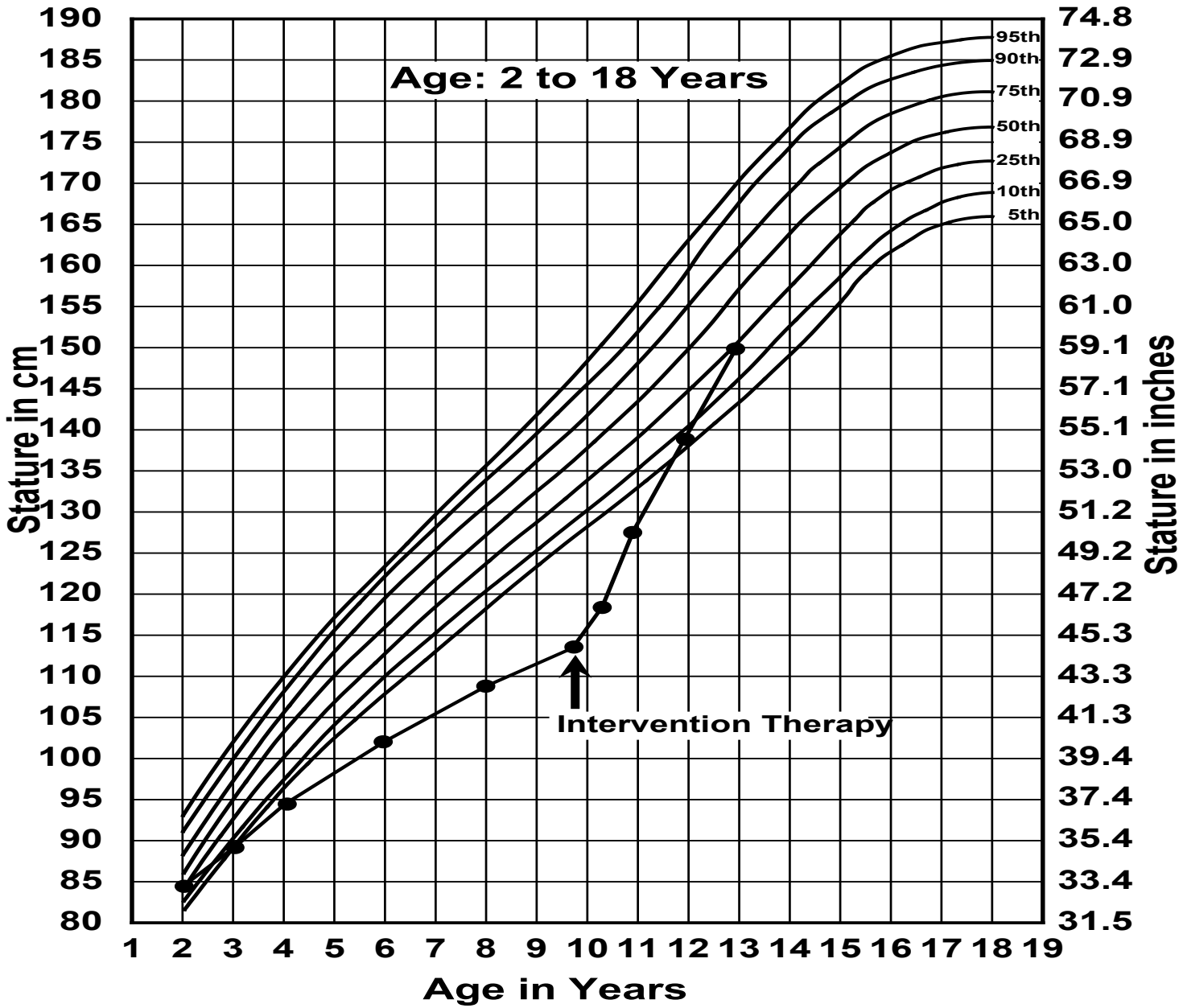
Renal Solute Load

Feeding	Potential Renal Solute Load, mOsm/liter
Human Milk	93
Milk based formula	135
Isolated Soy protein based formula	165
Evaporated milk formula	260
Whole cow milk	308

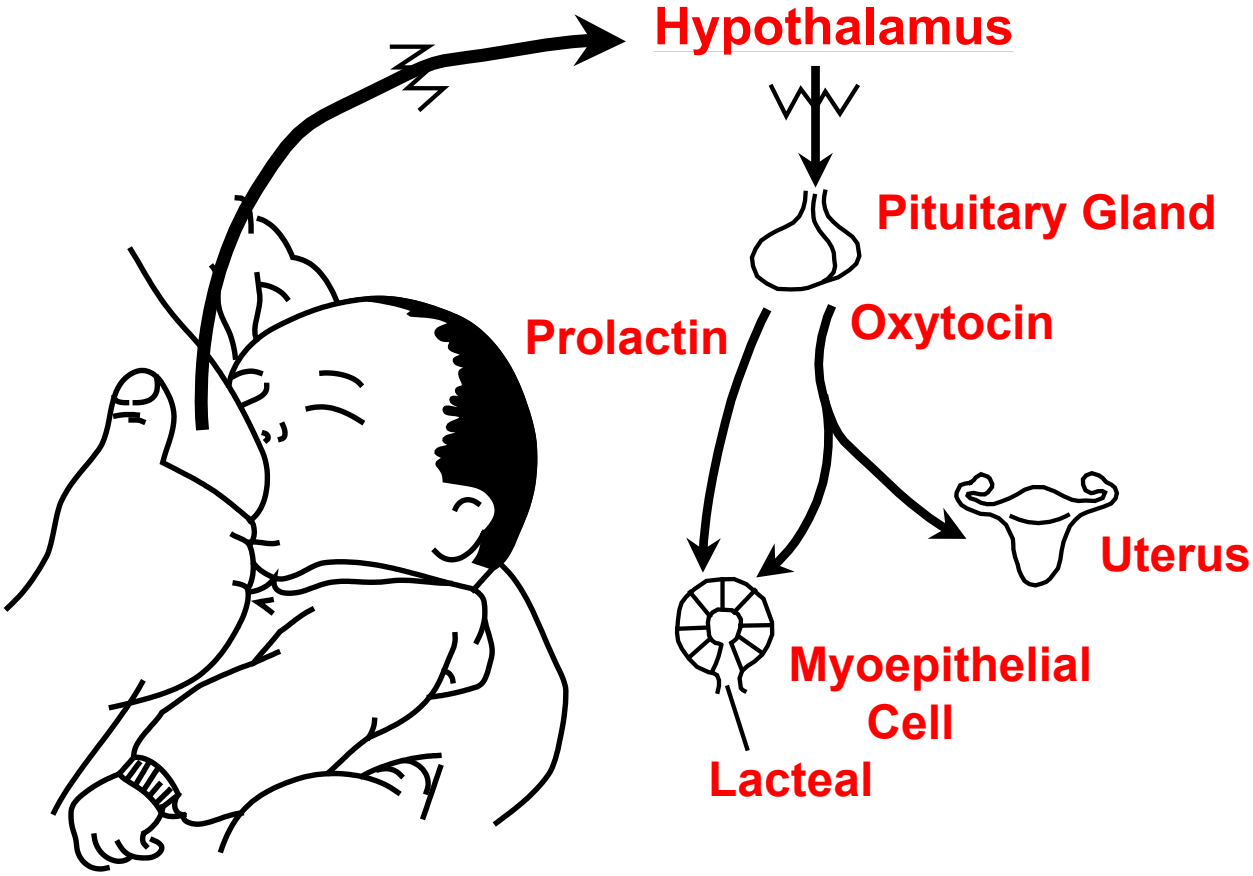
Failure to thrive due to increased calorie requirements

- Chronic / recurrent infection – urinary tract infection, tuberculosis
- Chronic respiratory insufficiency – bronchopulmonary dysplasia, CF
- Congenital / acquired heart disease
- Malignancy
- Chronic anemia
- Toxins – lead
- Drugs excess-thyroxine
- Endocrine disorrders - hyperthyroidism

Growth Chart
Boys
Stature for Age



BREAST FEEDING



Breastfeeding is the best

World Health Organization

Breast Milk Contents

Nutrients - Energy (67 Kcal/100ml)

CHO - lactose, oligosaccharides

Proteins - LF, α-lactalbumin,

β-casein, lysozyme;

cytokines, antibodies..

Fat - LC-PUFA (DHA, AA, EPA)

Micro-nutrients & trace metals

Cells - macrophages, neutrophils, lymphocytes

Breast milk contents

DID YOU EVER WONDER WHAT'S IN... ?

超過400種有益成分是奶粉裡没有的:
容易消化, 包括大量抗體、活細胞, 成長因子、荷爾蒙、酵素等等
+ 促進嬰兒發展的成分，如必需脂肪酸、DHA、 AA等。各成分互相配合

Breast milk is best ever treasure bestowed to babies by his/her mothers – the best food for best growth and development

BREASTMILK

WATER
CARBOHYD
Lactose
Oligosac
CARBOXYL
Alpha hyd
Lactic a
PROTEINS (building m
Whey pro
Alpha-l
HAM
Made
Lactofe
Many a
Casein
Serum alb
NON-PROT
Creatine
Creatinine
Urea
Uric acid
Peptides
Amino Ac
Alanine
Arginin
Aspartate
Glycine
Cystine
Glutamate
Histidine
Isoleucine
Leucine
Lycine
Methionine
Phenylalanine
Proline
Serine
Taurine
Theronine
Tryptopha
Tyrosine
Valine
Carnitine
make use
Nucleotides
the structura
5'-Adenos
3'5'-Cyclic
(3'5'-cyclic
5'-Cytidine
Cytidine di
Guanosine
Guanosine
3'- Uridine
5'-Uridine
Uridine diphosphate (UDP)
Uridine diphosphate hexose (UDPH)
Uridine diphosphate-N-acetyl-hexosamine

FATS
Sphingolipids
Gangliosides
GM1
GM2
GM3
Glucosylceramide
Glycosphingolipids
Galactosylceramide
Lactosylceramide
Globotriaosylceramide (GB3)
Globoside (GB4)
Sterols

VITAMINS
Iodine
Selenium
Choline
Sulpher
Chromium
Cobalt
Fluorine
Nickel
METAL
Molybdenum (essential element in many enzymes)

Epithelial growth factor (EGF)
Transforming growth factor-α (TGF-α)
TGF β1

PEPTIDES
Glycogen
Insulin
Corticosterone
Thrombopoietin
Gonadotropin-releasing hormone (GnRH)
GRH
Leptin (aids in regulation of food intake)
Ghrelin (aids in regulation of food intake)
Adiponectin
Feedback inhibitor of lactation (FIL)
Eicosanoids
Prostaglandins (enzymatically derived from

Phosphatase
Xanthine oxidase

ANTIPROTEASES
Complement C3
Complement C4
Complement C5
Complement C6
Complement C7
Complement C8
Complement C9
Glycoproteins
Mucins (attaches to bacteria and viruses to prevent them from clinging to mucousal tissues)
Lactadherin
Alpha-lactoglobulin

FORMULA

WATER
CARBOHYDRATES
Lactose
Corn maltodextrin
PROTEIN
Partially hydrolyzed reduced minerals whey protein concentrate (from cow's milk)
FATS
Palm olein
Soybean oil
Coconut oil
High oleic safflower oil (or sunflower oil)
M. alpina oil (Fungal DHA)
C.cohnii oil (Algal ARA)
MINERALS
Potassium citrate
Potassium phosphate
Calcium chloride
Tricalcium phosphate
Sodium citrate
Magnesium chloride
Ferrous sulphate
Zinc sulphate
Sodium chloride
Copper sulphate
Potassium iodide
Manganese sulphate
Sodium selenate
VITAMINS
Sodium ascorbate
Inositol
Choline bitartrate
Alpha-tocopheryl acetate
Niacinamide
Calcium pantothenate
Riboflavin
Vitamin A acetate
Pyridoxine hydrochloride
Thiamine mononitrate
Folic acid
Phylloquinone
Biotin
Vitamin D3
Vitamin B12
ENZYME
Trypsin
AMINO ACID
Taurine
L-Carnitine (a combination of two different amino acids)
NUCLEOTIDES
Cytidine 5-monophosphate
Disodium uridine 5-monophosphate
Adenosine 5-monophosphate
Disodium guanosine 5-monophosphate
Soy Lecithin

October 2012 | unicef UNITED KINGDOM

Preventing disease and saving resources: the potential contribution of increasing breastfeeding rates in the UK

Long-term effects of breastfeeding

A SYSTEMATIC REVIEW

Bernardo L. Horta, MD, PhD
Universidade Federal de Pelotas, Pelotas, Brazil

Cesar G. Victora, MD, PhD
Universidade Federal de Pelotas, Pelotas, Brazil

Short-term effects of breastfeeding

A SYSTEMATIC REVIEW
ON THE BENEFITS OF BREASTFEEDING
ON DIARRHOEA AND
PNEUMONIA MORTALITY

Bernardo L. Horta, MD, PhD
Universidade Federal de Pelotas, Pelotas, Brazil

The Surgeon General's Call to Action to Support Breastfeeding

2011

U.S. Department of Health and Human Services

世界衛生組織 2013 報告

© World Health Organization 2013

World Health Organization

Advantages of Breast Feeding

Benefits to Baby

- Nutritional value: best composition with high bioavailablity
- Reduced obesity and overfeeding
- Protect against infection & allergy
- Less contamination, readily available
- Enzymes, hormones and immune factors

Offers unique immunologic protection

- matches with the sequence of postnatal development of the immune system

- help adaptation of the gastrointestinal tract in the switch from fetal to postnatal life

Protective properties of breastfeeding

Three overlapping groups of bioactive Agents:

- **Direct acting antimicrobial agents**
- **Anti-inflammatory agents**
- **Immunomodulating agents**

Protection against infections & atopy

Increased health risks of not breastfeeding (baby)

- Diabetes 40%
- Obesity 25%
- Recurrent ear infection 60%
- Hospitalisaton for asthma or pneumonia 250%
- Death in first year 27%

Advantages of Breast Feeding

Benefits to Mother

- Involution of uterus
- Better physical shape
- Reduce neoplasm
- Improves psychological well being
- Less postnatal depression

Increased health risks of not breastfeeding (mother)

- Maternal Breast cancer 39%
- Maternal Type 2 diabetes 14%/yr

Advantages of Breast Feeding

Benefits to Family

- Maternal-infant bonding (attachment) (less school withdrawal, behavioral problem and abuse of child)
- Contraceptive effect (for birth control)
- Most economic & effective way of feeding

Advantages of Breast Feeding

Benefits to the Society

- Less medical consultations
- Less hospitalizations
- Less medical expense related to infections

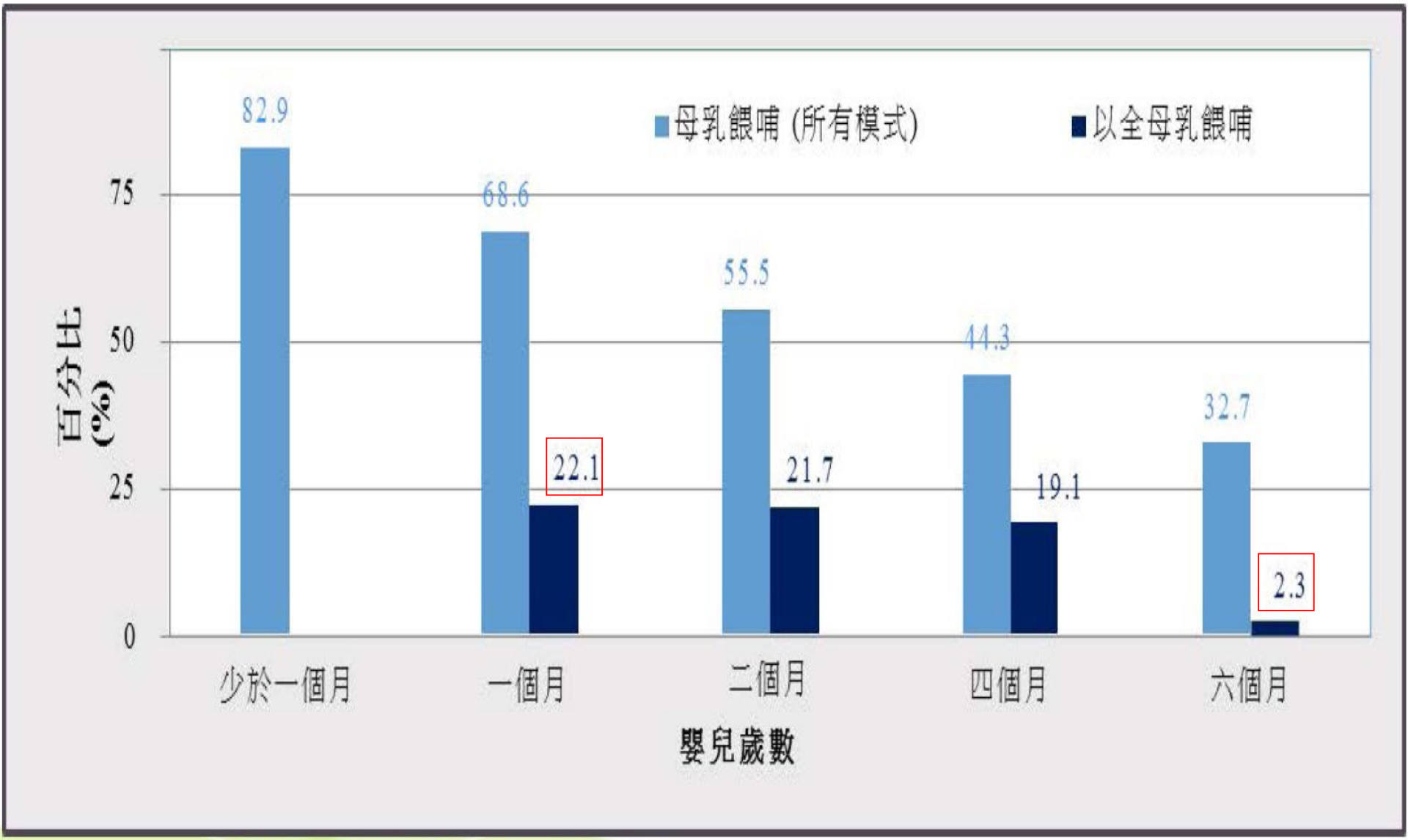
經濟效益 - 全母乳喂養4-6 月, 每年可節省

	美國(US$)	英國(£)
中耳炎	766 M	0.7 -1.2 M
腹瀉	162 M	3.6 -5.5 M
壞死性腸炎	2438 M	2.3-9.9 M
下呼吸道感染	1929 M	6.7 -11 M
哮喘	377 M	
白血病	134 M	
肥胖	404 M	15-28 M
糖尿病	70M	
濕疹	497 M	
嬰兒猝死綜合症	3722 M	4.7 M
乳癌		31 M
卵巢癌		
總計	> 100億美元	
智力 +3		

Breast feeding - disadvantages

- Physical exhaustion of mother (frequent, on demand feed)
- Emotional stress on mother
- Sleeping quality (mother) impaired
- Infections – virus e.g. HIV, CMV, HTLV
- Transmission of undesirable drugs e.g.Chemo-, radiotherapy, psychiatric drugs
- Inborn error of metabolism – *special diet*

圖二：於二零一二年出生的嬰兒以母乳餵哺的模式



資料來源：衞生署家庭健康服務。

表一：選定國家及地區的母乳餵哺率 (%)

	曾以母乳餵哺的比率 (%)	以全母乳餵哺的比率 (%)			
		一個月	兩個月	四個月	六個月
挪威 (2010)	98.5	84	75	48	10
澳洲 (2012)	96	-	-	39	15
英國 (2012)	81	-	-	12	1
美國 (2013)	76.5	48.6	42.5	29.1	16.4
台灣 (2013)	-	66.9	56.1	44.9	27.9
日本 (2011)	-	51.6	55.0	55.8	-
馬來西亞 (2010)	94.7	-	-	19.3	14.5
新加坡 (2013)	96	35	28	-	1
香港 (2012)	82.9	22.1	21.7	19.1	2.3

資料來源：衞生署家庭健康服務。

WHO Recommendation

- accurate information, and the support of their family, the health care system and society at large

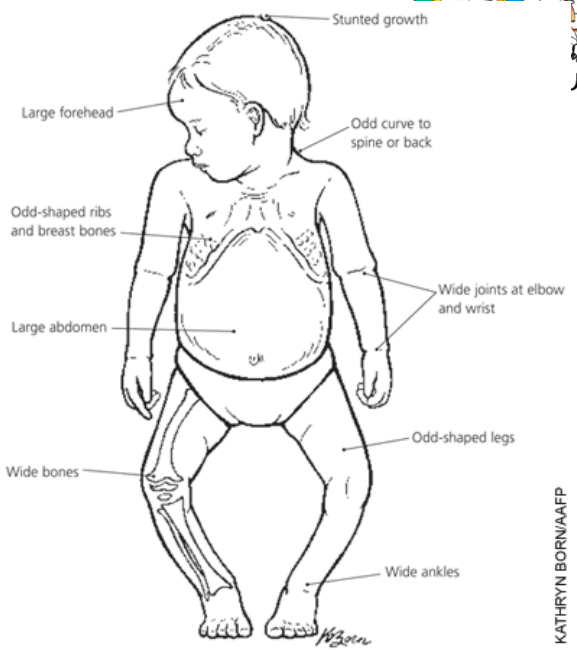
- Exclusive breastfeeding is recommended up to 6 months of age, with continued breastfeeding along with appropriate complementary foods up to 2 years of age or beyond

Vitamin D and its significance

- Key modulator on absorption and metabolism of different minerals
- Over **one billion** people globally have insufficient vitamin D
- Vitamin D deficiency during infancy is associated with
  - Growth retardation
  - Skeletal deformities
  - Adverse effects on the immune system
  - Brain development and psychiatric conditions

Rickets in children

- One of the most common health problem related to vitamin D deficiency
- A condition that affects bone development in children
- Can induce
  - bone pain
  - poor growth
  - Bone deformities



Source: Caruso and Fuzaylov (2013)

Stunting

- Impaired growth and developm that children experience from
  - Poor nutrition
  - Repeated infection
  - Inadequate psychosocial stimulation

Source: UNICEF

- Vitamin D deficiency was identified as one risk factor
  - Possible interaction with **growth hormone (GH)** and **insulin-like growth factor 1 (IGF-1)**

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journal homepage: www.nutritionjrnl.com

Applied nutritional investigation

Low intake of calcium and vitamin D, but not zinc, iron or vitamin A, is associated with stunting in 2- to 5-year-old children

Martha E. van Stuijvenberg Ph.D. [a,b,*], Jana Nel M.Nutr. [c], Serina E. Schoeman M.Cur. [a], Carl J. Lombard Ph.D. [d], Lisanne M. du Plessis M.Nutr. [e], Muhammad A. Dhansay F.C.Paed. [a,e,f]

Allergies

Allergy EUROPEAN JOURNAL OF ALLERGY AND CLINICAL IMMUNOLOGY

REVIEW ARTICLE | Free Access

Asthma, allergy and respiratory infections: the vitamin D hypothesis

S. Bozzetto, S. Carraro, G. Giordano, A. Boner, E. Baraldi

Source: Lodge et al. (2020)

- Vitamin D as an **important modulator** on the innate and adaptive immune responses
- Serum Vitamin D level ↓ ➔ Disease severity ↑
- Vitamin D deficiency is common among autoimmune disease patients
  - Systemic lupus erythematosus (SLE), Multiple sclerosis (MS), Rheumatoid arthritis (RA)

Local large-scale study

Primary objective

- To estimate the prevalence of vitamin D insufficiency and deficiency in pregnant women, infants and toddlers

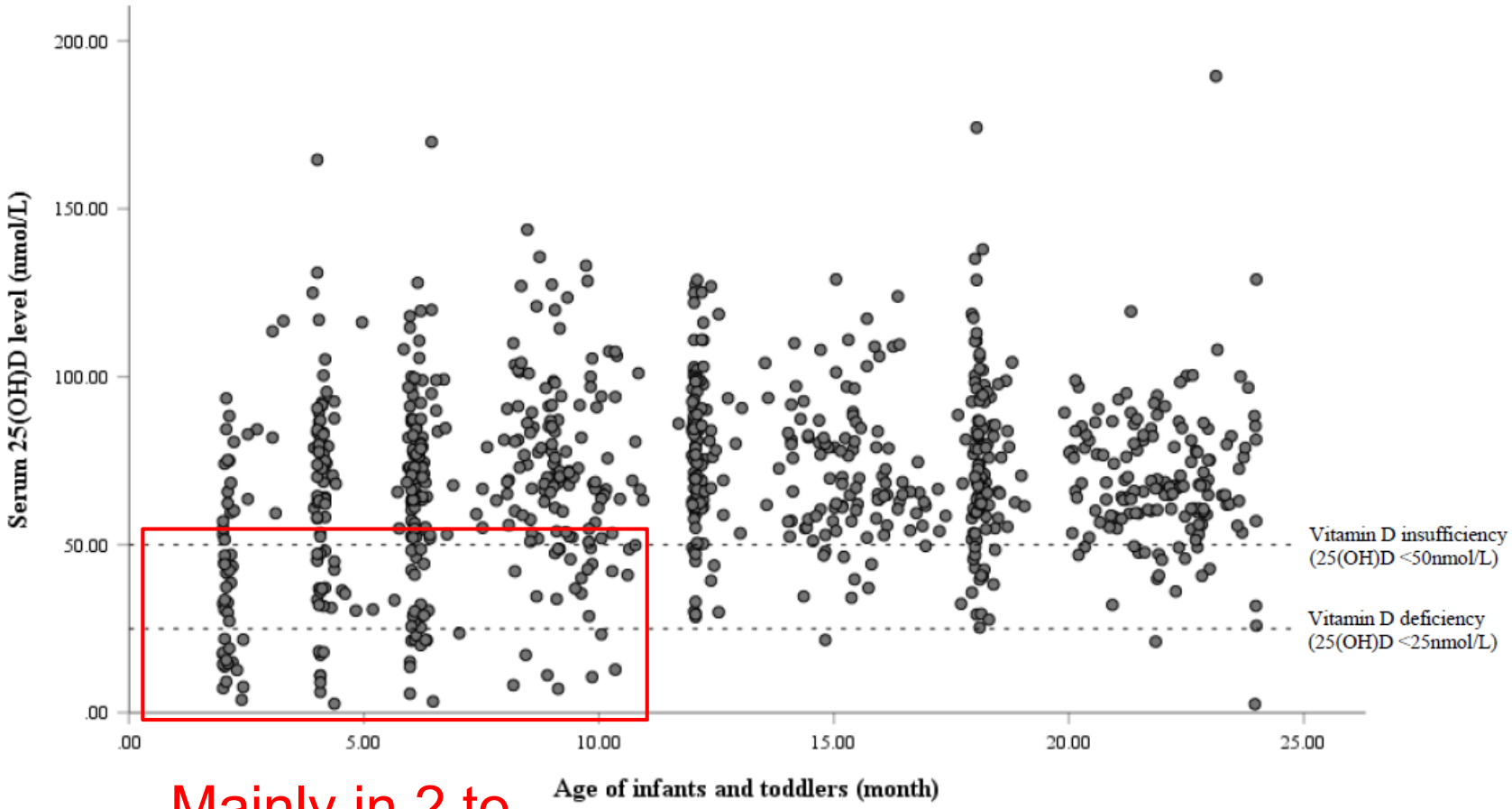
Secondary objective

- To investigate the effect of various factors on the vitamin D status of pregnant women, infants and toddlers
  - Mode of infant feeding
  - Dietary intake
  - Supplementation
  - Sunlight exposure

	serum total 25(OH)D concentration[1,2]
Sufficiency	>50 nmol/L
Insufficiency	25 - 50 nmol/L
Deficiency	<25 nmol/L

Vitamin D status of infants and toddlers

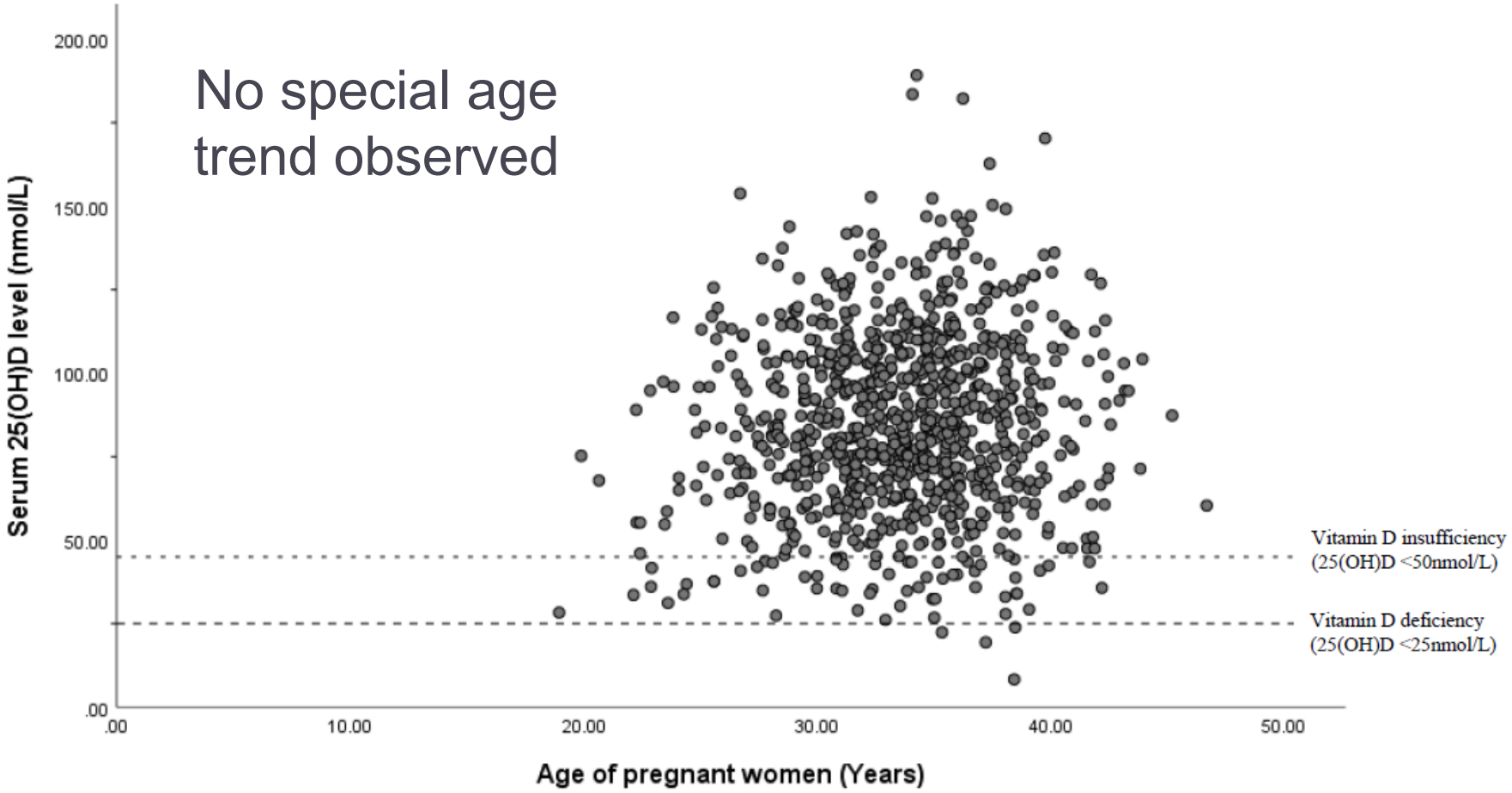
Figure 1. Vitamin D level of enrolled infants and toddlers



Mainly in 2 to 10 months

Vitamin D status of pregnant women

Figure 2. Vitamin D level of enrolled pregnant women



UPAM

nutrients

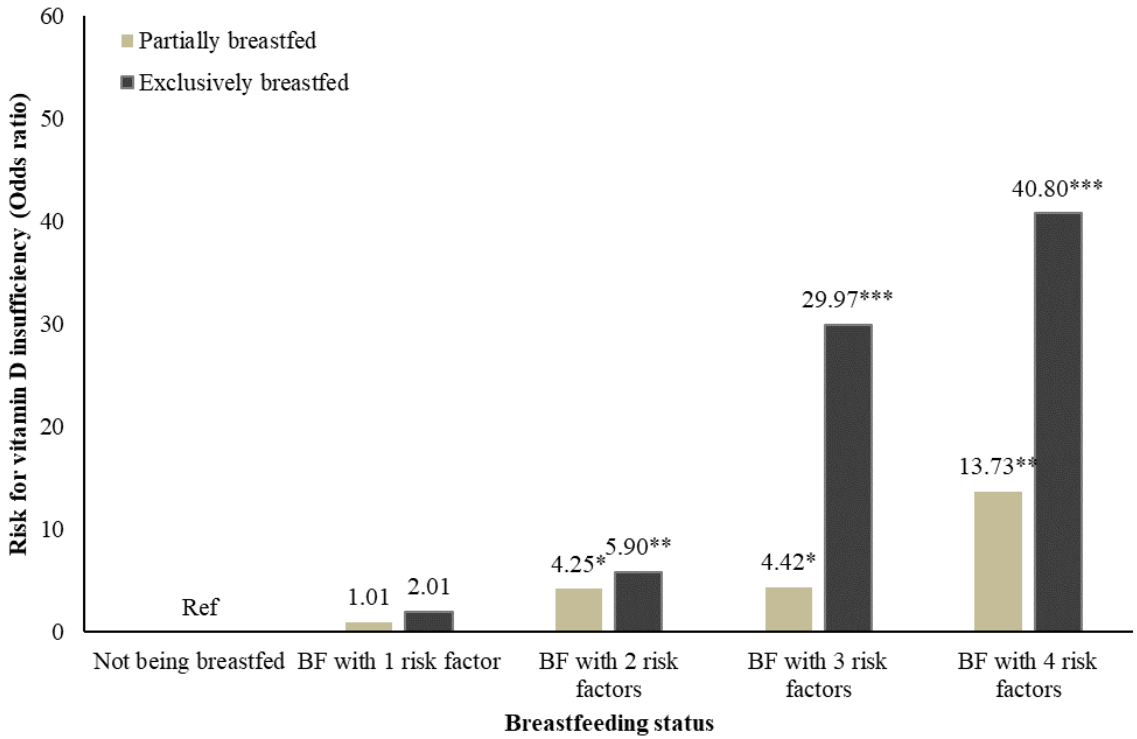
Article

An Assessment of Risk Factors for Insufficient Levels of Vitamin D during Early Infancy

Keith T. S. Tung [1], Rosa S. Wong [1], Hing Wai Tsang [1], Bianca N. K. Chan [1], Siew Yan Wong [1], Hung-Kwan So [1], Joanna Y. L. Tung [1,2], Marco H. K. Ho [1], Wilfred H. S. Wong [1] and Patrick Ip [1,*]

- This study examined the risk factors for vitamin D insufficiency in early infancy (2 to 6 months)
- Cumulative risk approach was used
- Risk factors examined: the infant being a girl, low family income, having a multiparous mother, no vitamin supplementation, sun cream being applied by their parents, and a mother with smoking history

Dose–response pattern

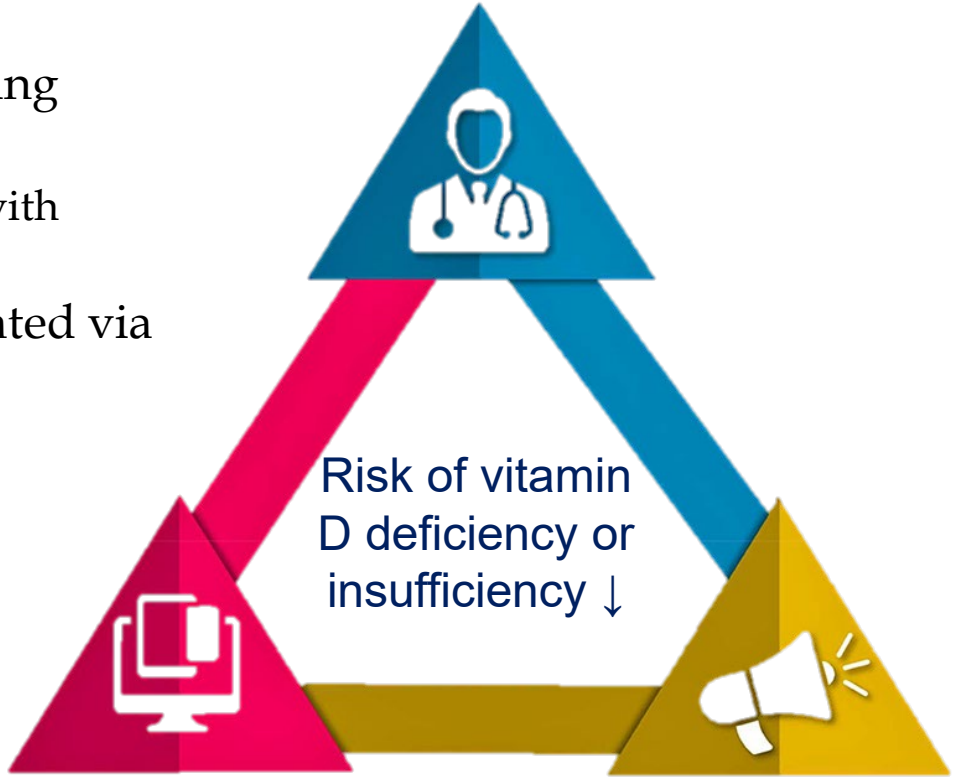


Risk factors examined: the infant being a girl, low family income, having a multiparous mother, no vitamin supplementation, sun cream being applied by their parents, and a mother with smoking history

- Occurrence of multiple risk factors is the main reason for higher risk of vitamin D insufficiency among breastfed infants

Recommendation

- Adequate support and guidance should be provided by clinicians
  - Information on the importance of vitamin D
  - Provision of supplementation
  - Education over the source of vitamin D intake (E.g. regular sunlight exposure, type of food with vitamin D)
- Clinicians should help in identifying high-risk groups
  - E.g. Exclusively breastfed infants with no regular supplementation
- Information can also be disseminated via
  - Social media / mass media
  - Mobile app / e-copy of information leaflet



Take home message

1) Definition of stunting

2) Common causes of failure to thrive

3) Benefits of breast feeding

4) Importance of vitamin D

Are we too thin?

Thank You!

The evaluation questionnaire(s) for the MBBS Whole Class Session(s)

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